

Hellenic Complex Systems Laboratory

# Relation of Positive and Negative Predictive Values of Diagnostic Tests

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# Relation of Positive and Negative Predictive Values of Diagnostic Tests

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## Short Description of the Demonstration

This Demonstration examines the relation of the negative predictive value (NPV) and the positive predictive value (PPV) of a diagnostic test for normally distributed nondiseased and diseased populations. Different levels of disease prevalence are considered. The means and standard deviations of the populations are defined by the user and expressed in arbitrary units.

## Search Terms

sensitivity, specificity, diagnostic test, diagnostic accuracy, normal distribution, positive predictive value, negative predictive value

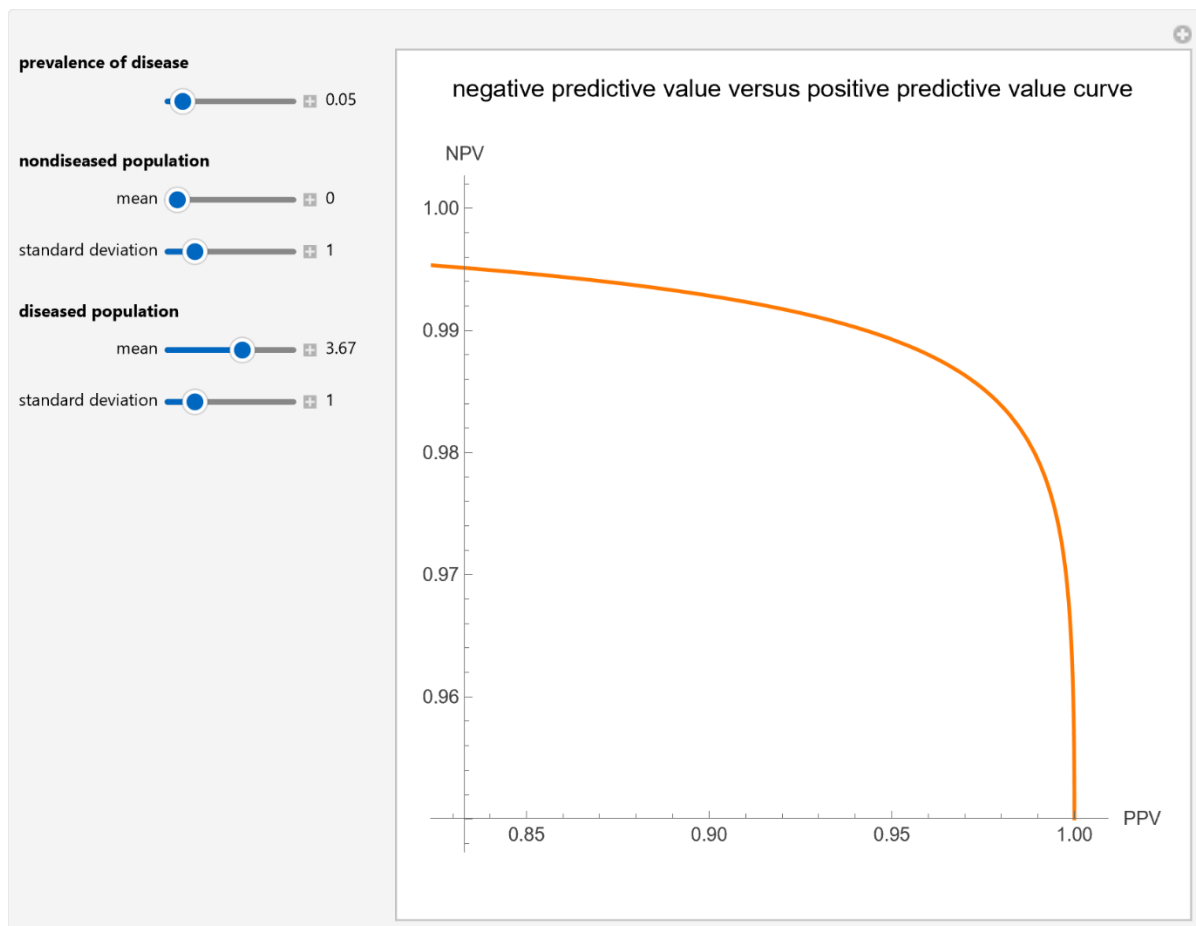


Figure 1: *Negative predictive value versus positive predictive value curve plot of a diagnostic test, with the settings shown on the left.*

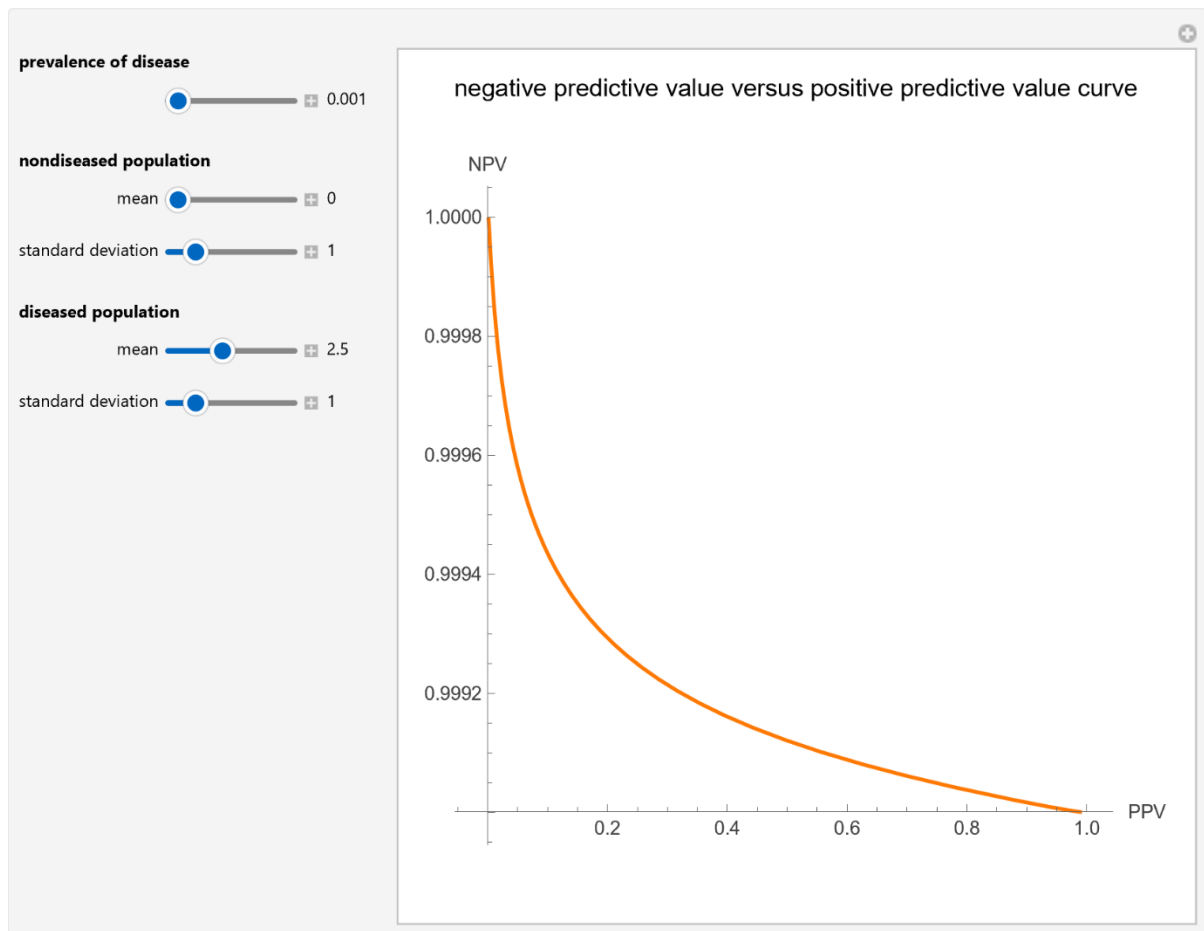


Figure 2: Negative predictive value versus positive predictive value curve plot of a diagnostic test, with the settings shown on the left.

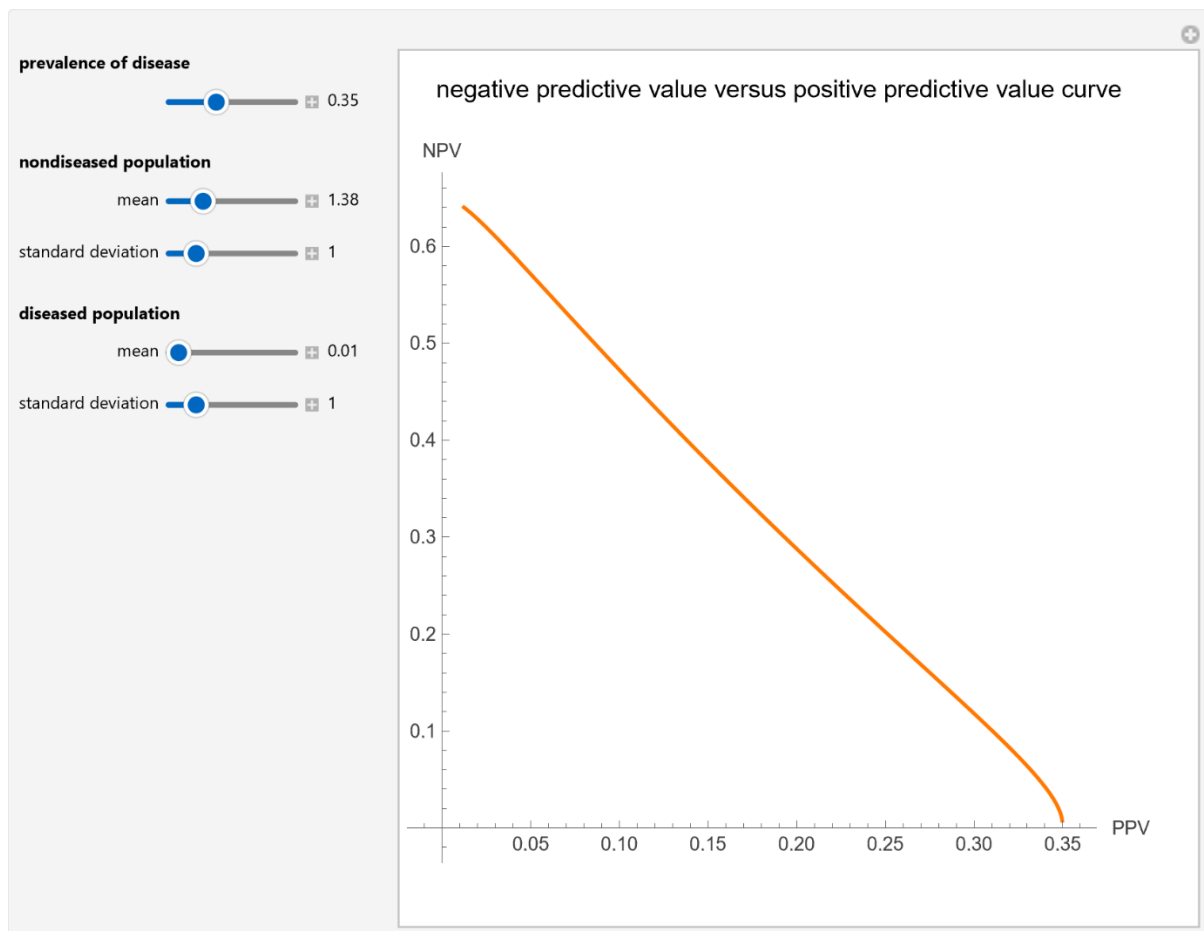


Figure 3: Negative predictive value versus positive predictive value curve plot of a diagnostic test, with the settings shown on the left.

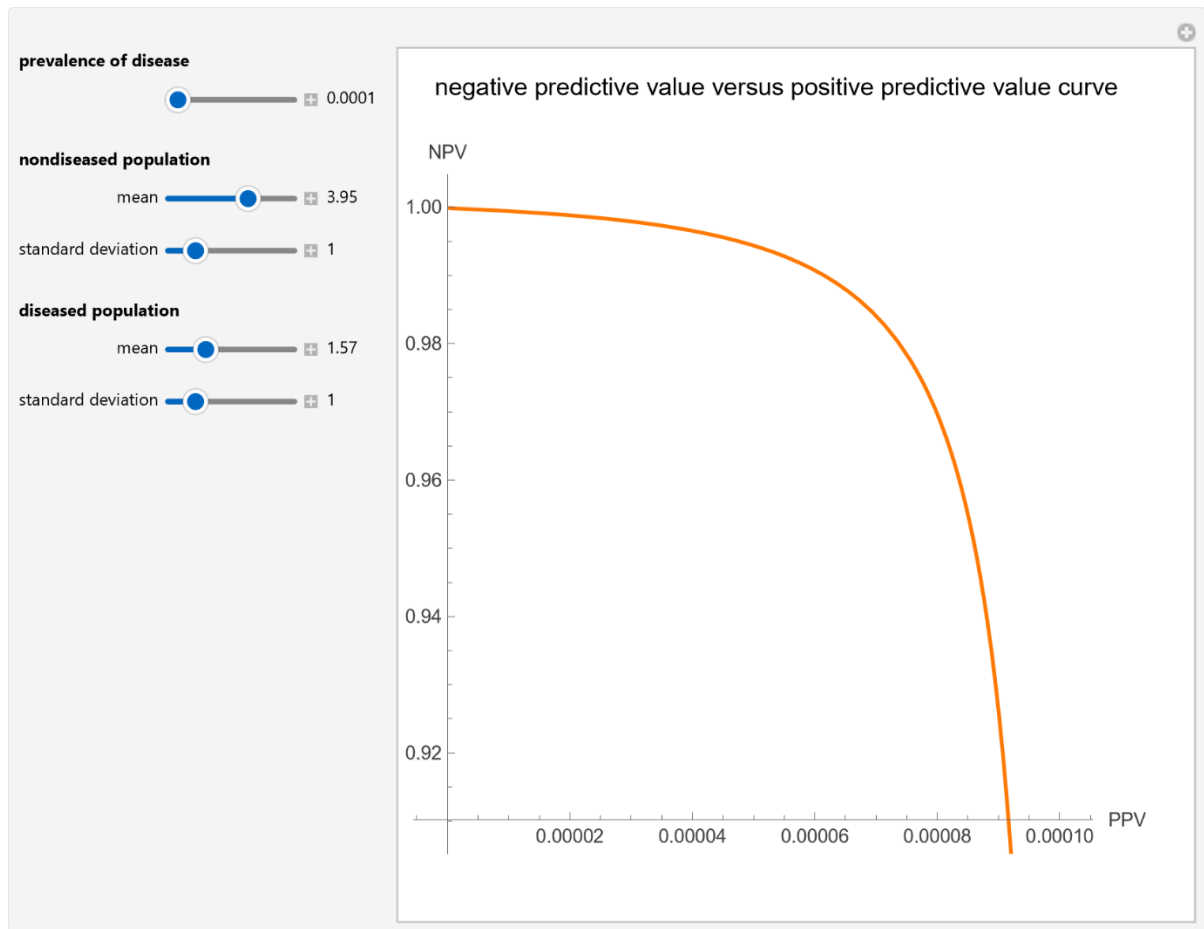


Figure 4: *Negative predictive value versus positive predictive value curve plot of a diagnostic test, with the settings shown on the left.*

## Details

The PPV and NPV that are used in the evaluation of the clinical accuracy of a diagnostic test applied to a diseased and nondiseased population can be calculated given the sensitivity and specificity of the test. Sensitivity is the fraction of the diseased population with a positive test result, while specificity is the fraction of the nondiseased population with a negative test result. If we denote by  $Se$  the sensitivity,  $Sp$  the specificity, and  $v$  the prevalence, then:

$$PPV = \frac{Se v}{Se v + (1 - Sp)(1 - v)}$$
$$NPV = \frac{Sp (1 - v)}{Sp (1 - v) + (1 - v) pr}$$

Given a diseased and a nondiseased population, the specificity can be defined as a function of sensitivity; therefore, we can plot NPV versus PPV (a parametric plot).

The population data in Figure 1 describe a bimodal distribution of serum glucose measurements in nondiabetic and diabetic populations [1].

## References

[1] T.-O. Lim, R. Bakri, Z. Morad and M. A. Hamid. Bimodality in Blood Glucose Distribution: Is It Universal? *Diabetes Care*, **25**(12), 2002 pp. 2212–2217.

## Demonstration

### Version

1.1.2

### DOI

[10.5281/zenodo.20565731](https://doi.org/10.5281/zenodo.20565731)

### Source

Programming language: Wolfram Language

Availability: The updated source notebook is available at:

<https://www.hcsl.com/Tools/Demonstrations/RelationOfPositiveAndNegativePredictiveValuesOfDiagnostic.nb>

First published in 2015 as part of the [Wolfram Demonstrations Project](#).

## Software Requirements

Operating systems: Microsoft Windows, Linux, Apple macOS and iOS

Other software requirements: Wolfram Player®, freely available at: <https://www.wolfram.com/player/> or Wolfram Mathematica®.

## System Requirements

Processor: x86-64 compatible CPU.

System memory (RAM): 4 GB+ recommended.

## Citation

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# Technical Report

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